

CS & IT ENGINEERING



Theory of Computation

- ① Alphabet
- ② string, $\epsilon \Rightarrow$ zero length string
- ③ Language $\Rightarrow L = \{ \epsilon \} \Rightarrow$ finite language
 $L = \{ \} \Rightarrow$ empty language

DFA

Lecture No.- 02



By- Venkat sir

Recap of Previous Lecture



Topic

Introduction of TOC

Topic

THEORY OF COMPUTATION



Topic

Finite Automaton & Regular Languages.

Topic

Pushdown Automata & Context free Languages.

Topic

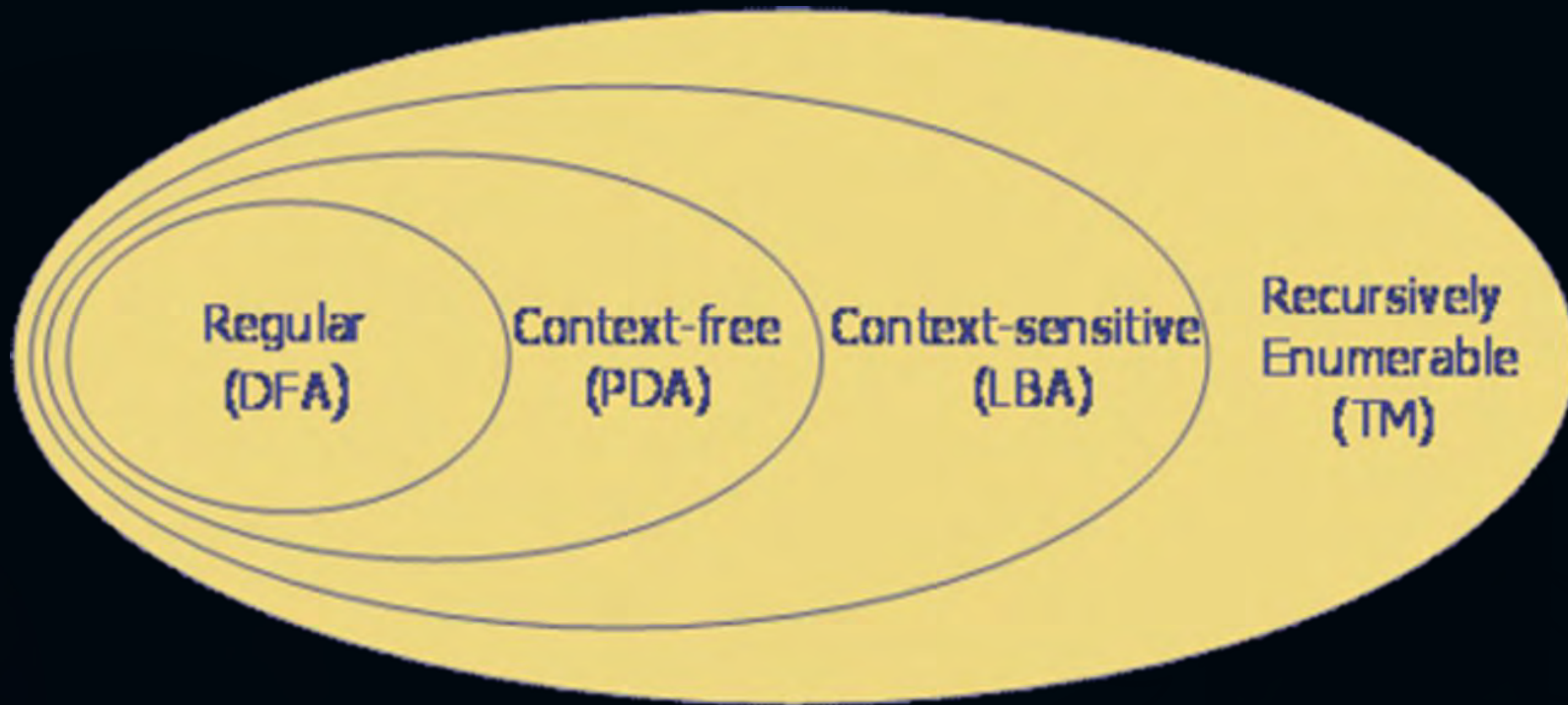
**Turing Machine &
Recursive Enumerable Languages.**

Topic

Undecidability.



Topic : Theory of Computation





Topic : Introduction:



It is the mathematical study of computing machines and their capability

or

It is the study of automata theory and formal languages.



Topic : Introduction:



Decidable Problem

Undecidable Problem



Topic : Terminologies:



Alphabet(Σ): Finite non-empty set of symbols

Ex:- $\{a, b\}$ –

$\{a, 1, 2\}$ –

$\{\}$ –



Topic : Terminologies:



Alphabet(Σ): Finite non-empty set of symbols

Ex:- $\{a, b\}$ –

$\{a, 1, 2\}$ –

$\{\}$ –



Topic : String:



String: Finite sequence of symbols over the given alphabet Σ .

Ex:-



Topic : String:

Language :- Any set of strings over the given alphabet $\Sigma = \{a, b\}$.

$$L_1 = \{ ab, ba, abab \} -$$

$$L_2 = \{ a, ab, aba, \} -$$

$$L_3 = \{ \} -$$

$$L_4 = \{ \epsilon \} -$$

$$L_5 = \{ a \} -$$

$$L_6 = \{ \epsilon, a, b, aa, ab, ba, bb, ----- \} -$$



Topic : String:



Sub – String : Consecutive sequence of symbols over the given string.

Total no of substring for the given string = $n(n + 1)/2 + 1$

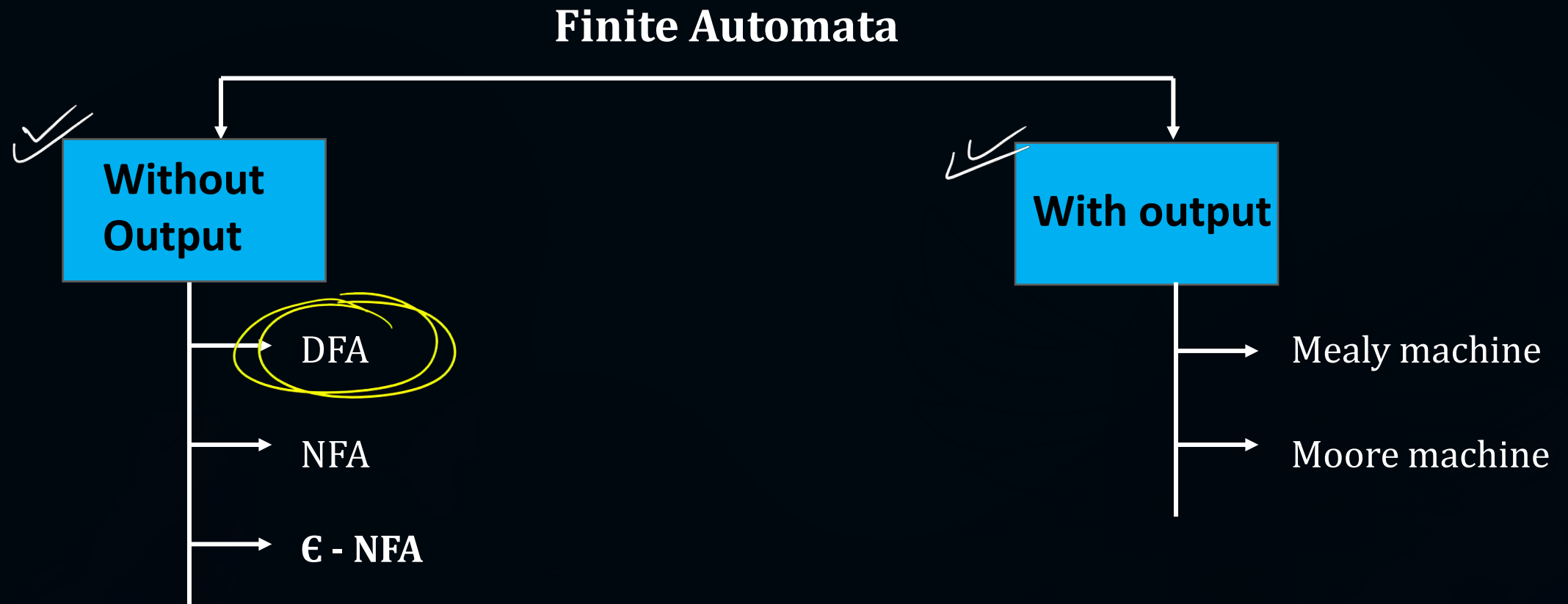


FINITE AUTOMATA



Topic : Finite Automata

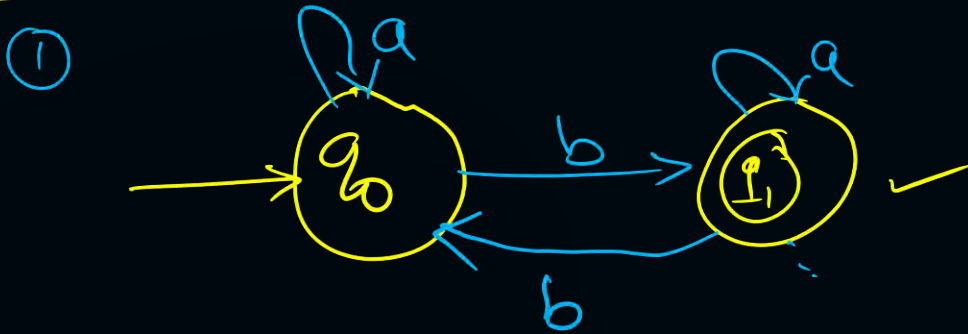
It is a mathematical model which contains finite number of states and transitions.





Topic : Deterministic Finite Automata

DFA : It is a finite automata in which from every state on every input symbol exactly one transition should exits.





Topic : Deterministic Finite Automata

FORMAL DFA :

Definition

DFA is defined as

$$\text{DFA} = \boxed{Q} \boxed{\Sigma} \boxed{q_0} \boxed{F} \boxed{\delta}$$

Q : Finite set of states

Σ : Input alphabet

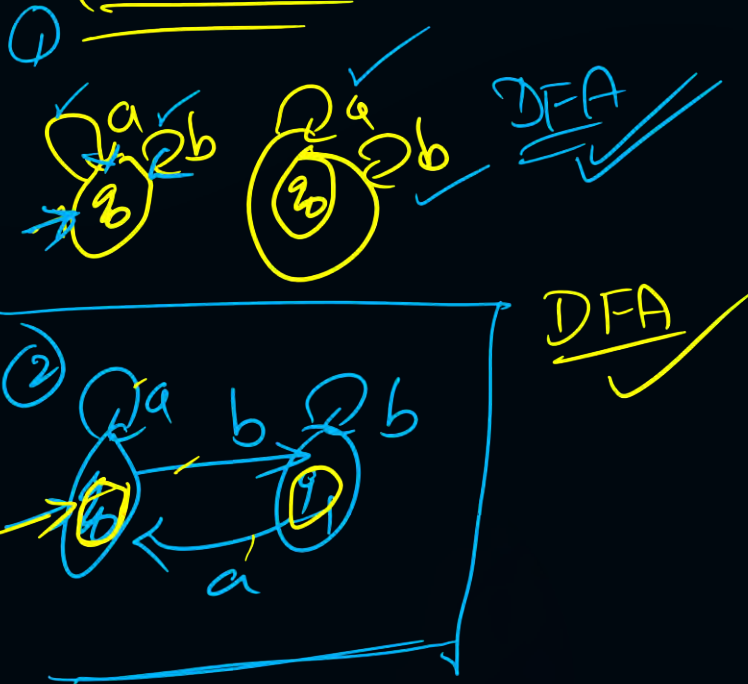
q_0 : Initial state $\rightarrow$ only one

F : Set of final states $\rightarrow$ any, 0, 1, 2, 3, 4

δ : Transition function $Q^* \Sigma \rightarrow Q$

$$\boxed{Q \times \Sigma \rightarrow Q}$$

0
1, 2, 3

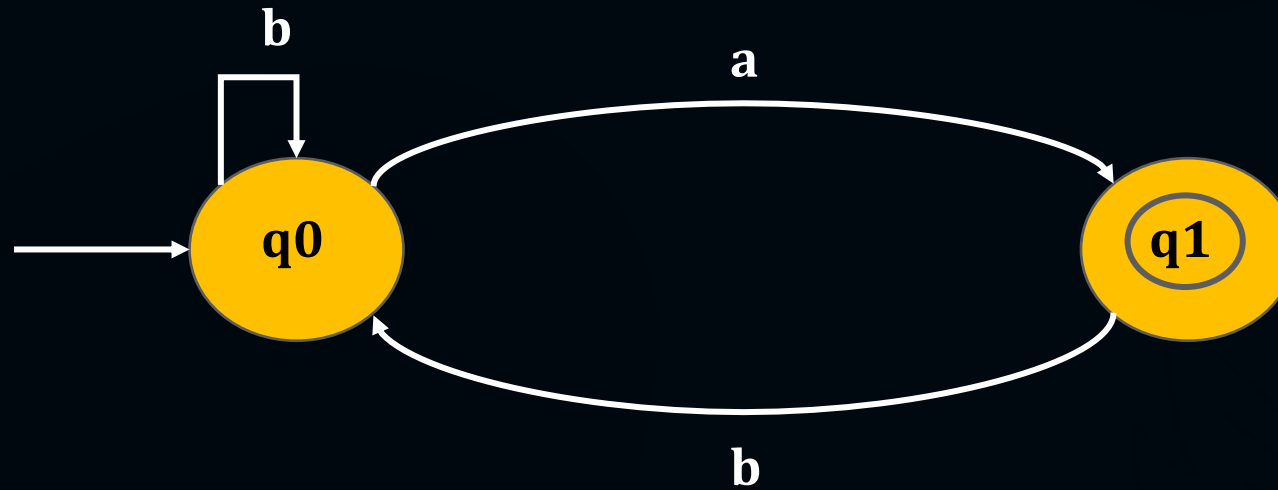




Topic : Deterministic Finite Automata

Example of DFA :

(1)

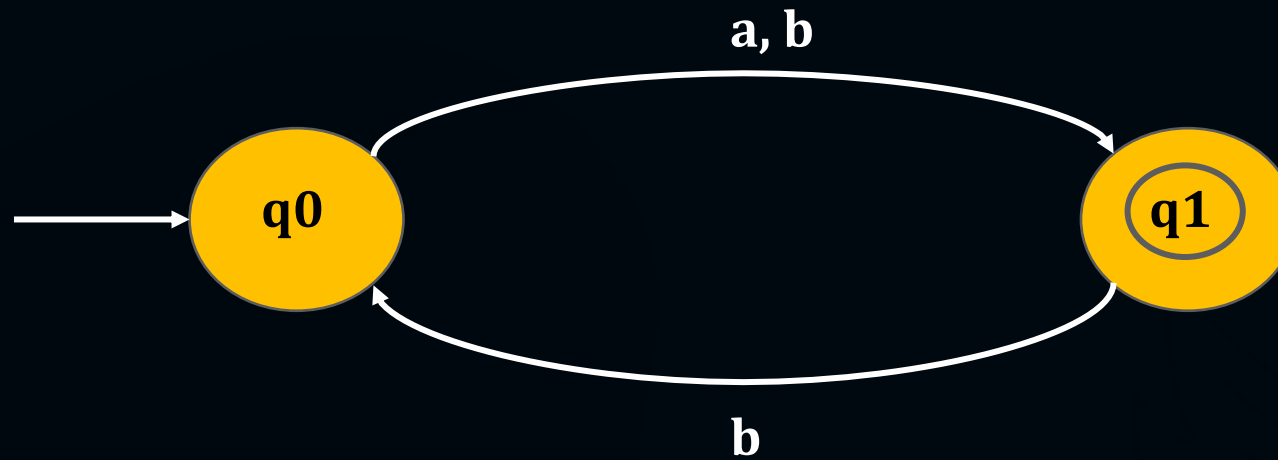




Topic : Deterministic Finite Automata

Example of DFA :

(2)

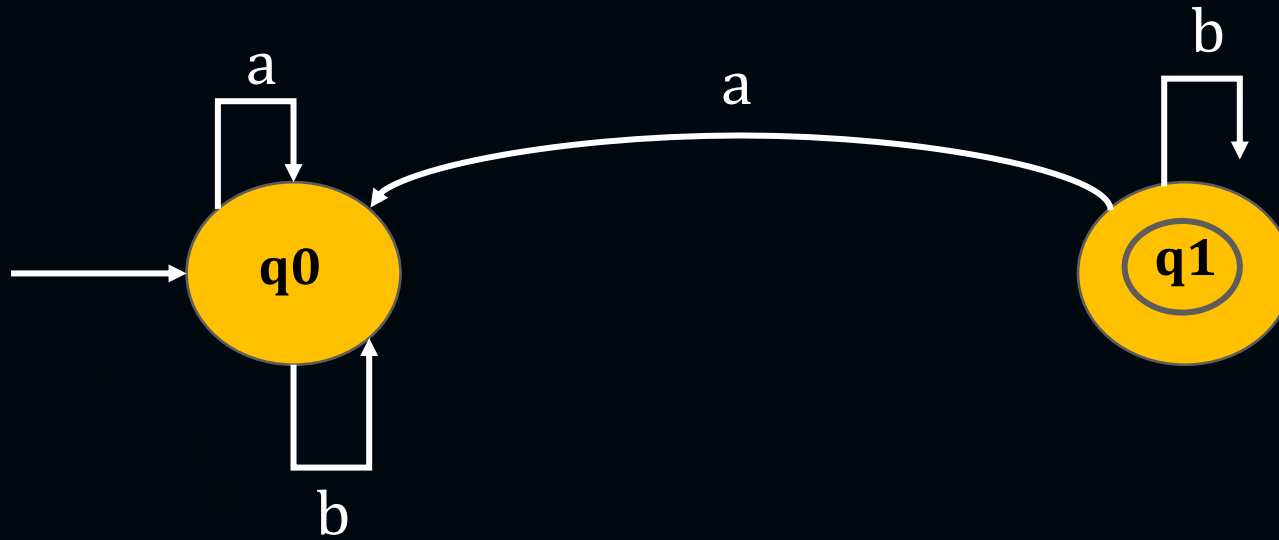




Topic : Deterministic Finite Automata

Example of DFA :

(3)

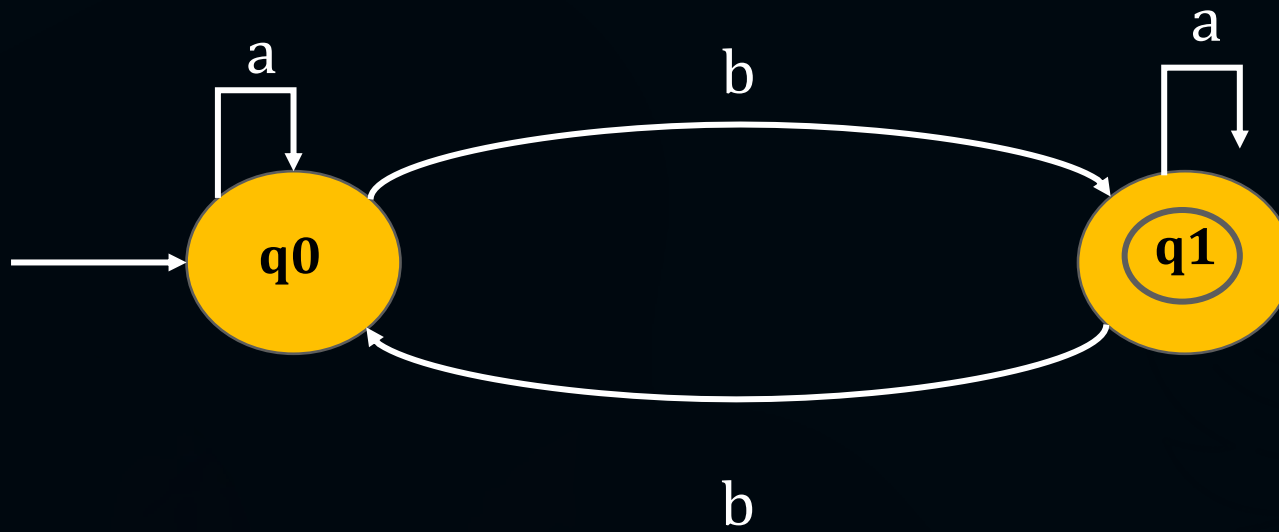




Topic : Deterministic Finite Automata

Example of DFA :

(4)





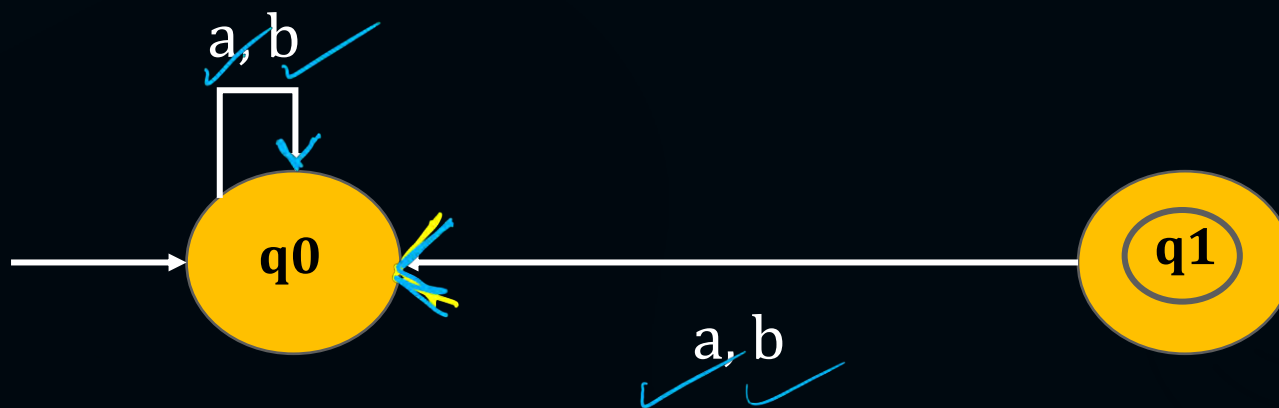
Topic : Deterministic Finite Automata

Example of DFA?

DFA

$(Q, \Sigma, q_0, F, \delta)$

(5)

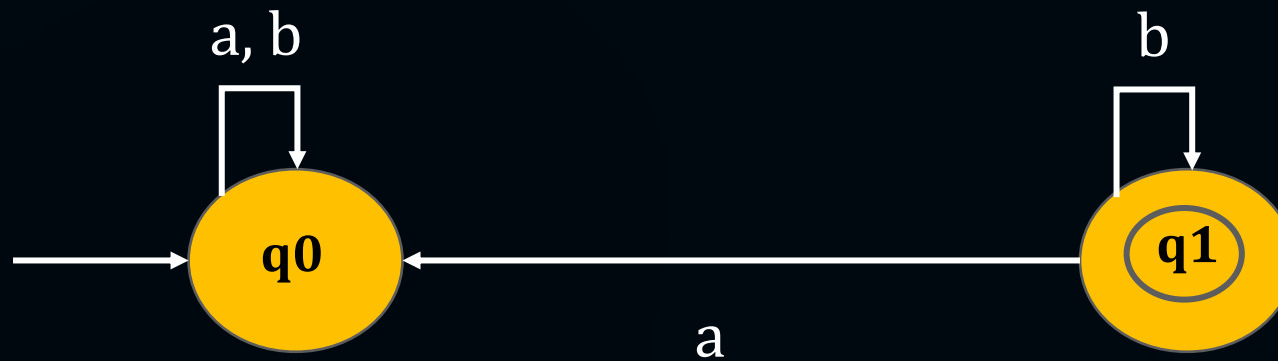




Topic : Deterministic Finite Automata

Example of DFA :

(6)

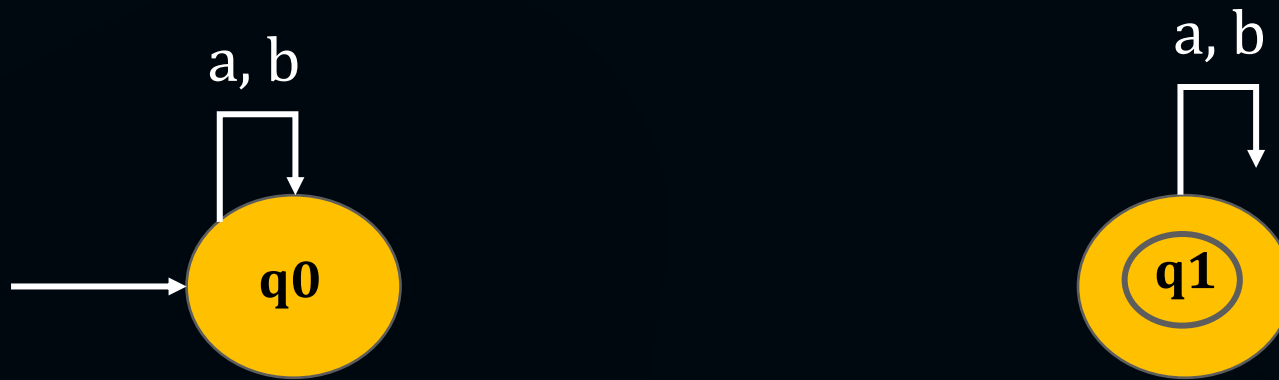




Topic : Deterministic Finite Automata

Example of DFA :

(7)

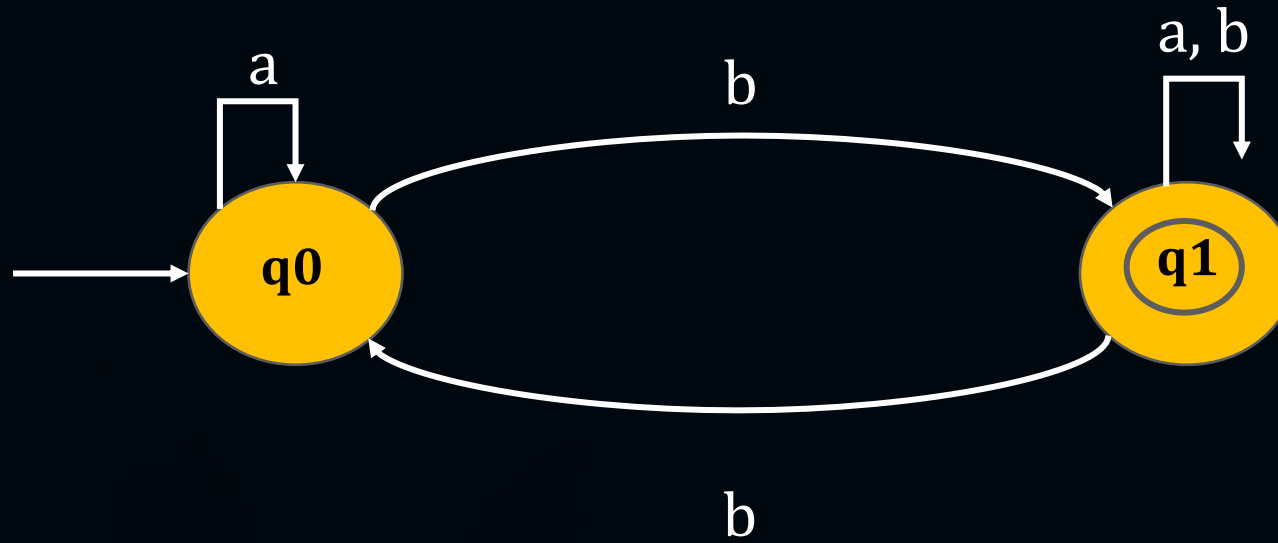




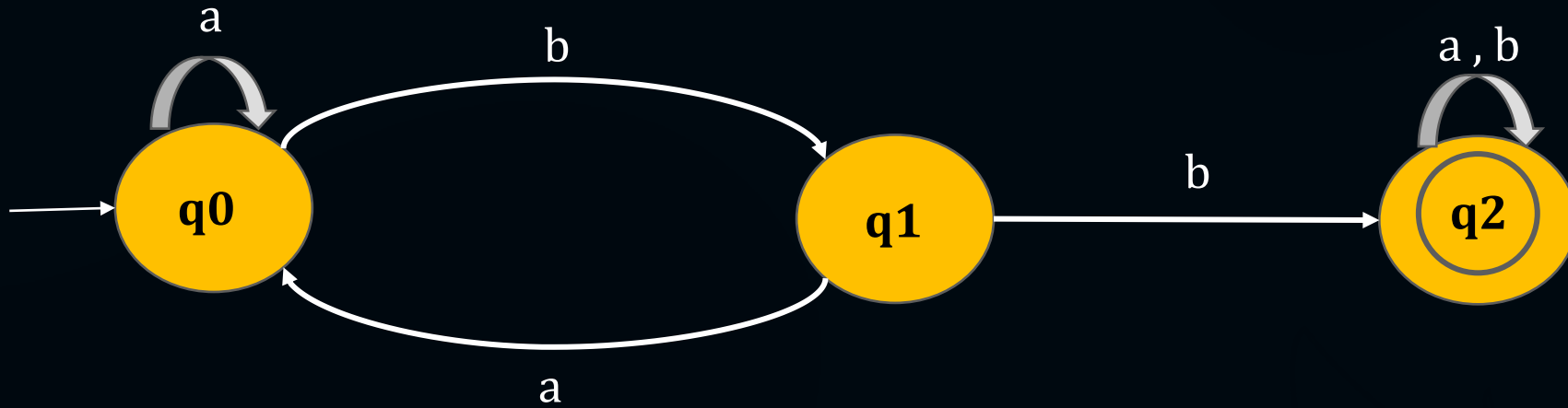
Topic : Deterministic Finite Automata

Example of DFA :

(8)



#Q. Identify language accepted by given DFA



Set of all strings

A

Starting with bb

B

Ending with bb

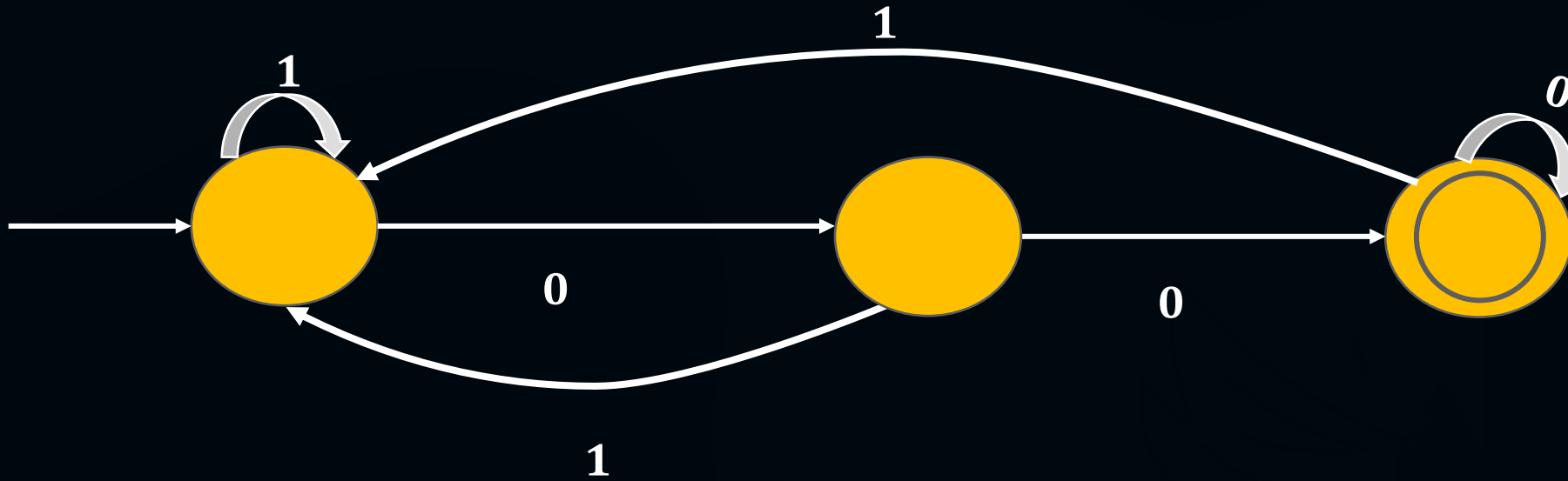
C

Contains at least 2 b's

D

None

#Q. Identify the language accepted by following DFA



A

Starting with 00

B

Substring 00

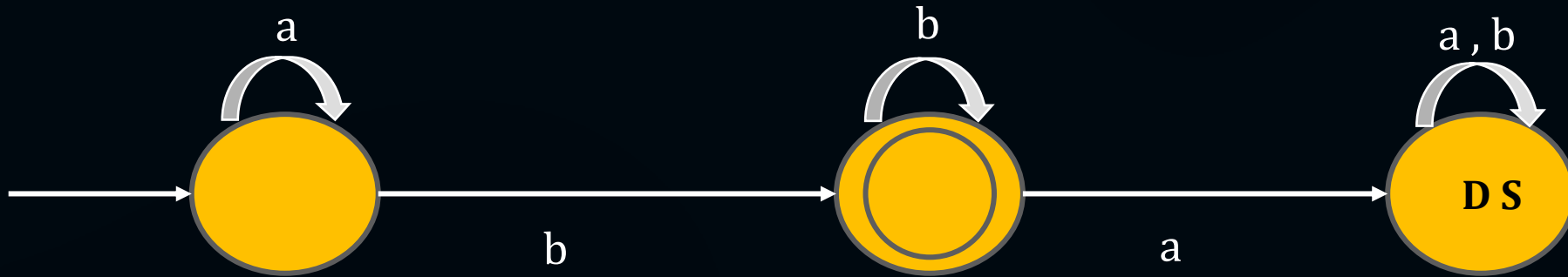
C

Ending with 0

D

None

#Q. Identify language accepted by following DFA



A

$$L = \{a^n b^m \mid n, m \geq 1\}$$

B

$$L = \{a^n b^m \mid n \geq 1, m \geq 0\}$$

C

$$L = \{a^n b^m \mid n, m \geq 0\}$$

D

None

#Q. Identify language accepted by following DFA



A

$$L = \{a^n b^m \mid n, m \geq 1\}$$

B

$$L = \{a^n b^m \mid n \geq 1, m \geq 0\}$$

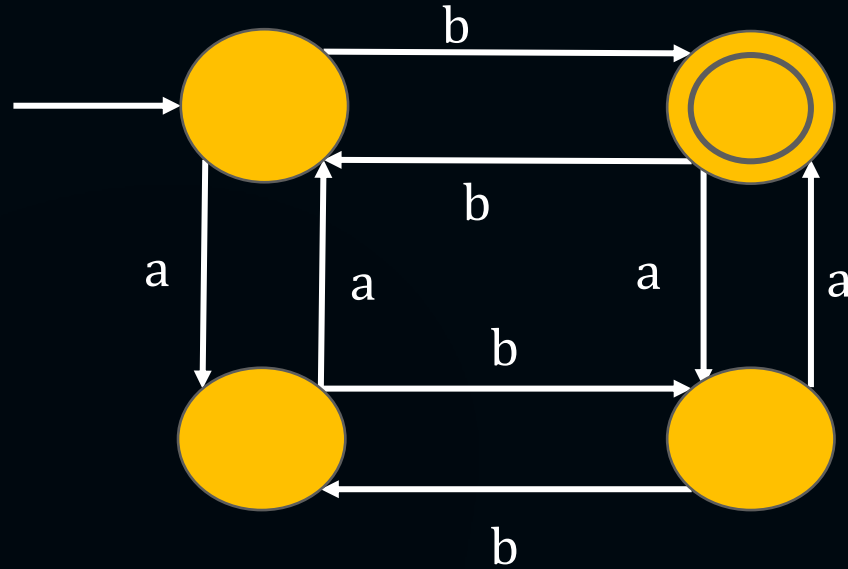
C

$$L = \{a^n b^m \mid n, m \geq 0\}$$

D

None

#Q. Identify language accepted by following DFA



A

a's even and # b's even

B

a's odd and # b's even

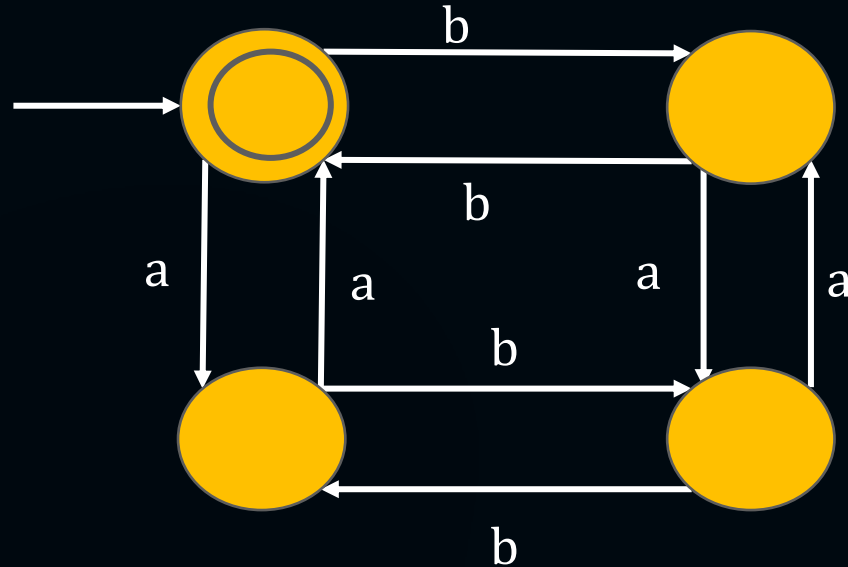
C

a's odd and # b's odd

D

a's even and # b's odd

#Q. Identify language accepted by following DFA



A

a's even and # b's even

B

a's odd and # b's even

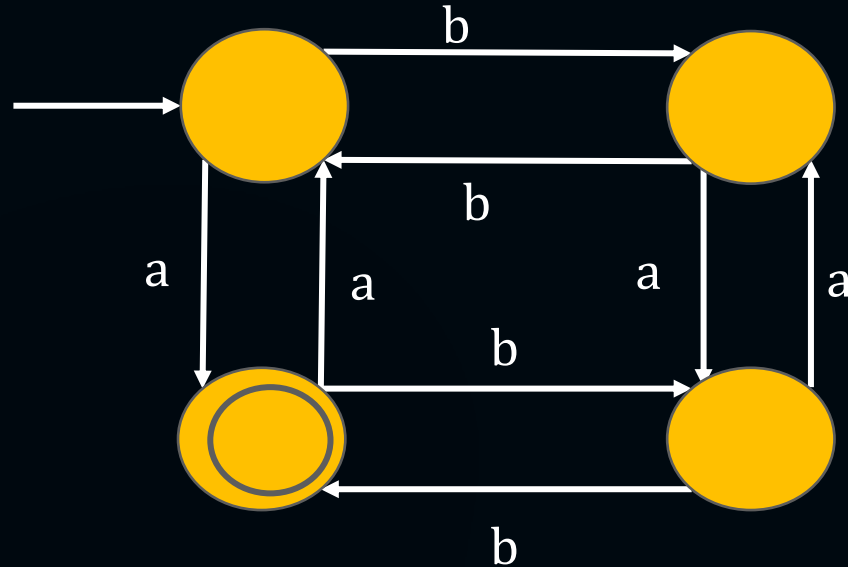
C

a's odd and # b's odd

D

a's even and # b's odd

#Q. Identify language accepted by following DFA



A

a's even and # b's even

B

a's odd and # b's even

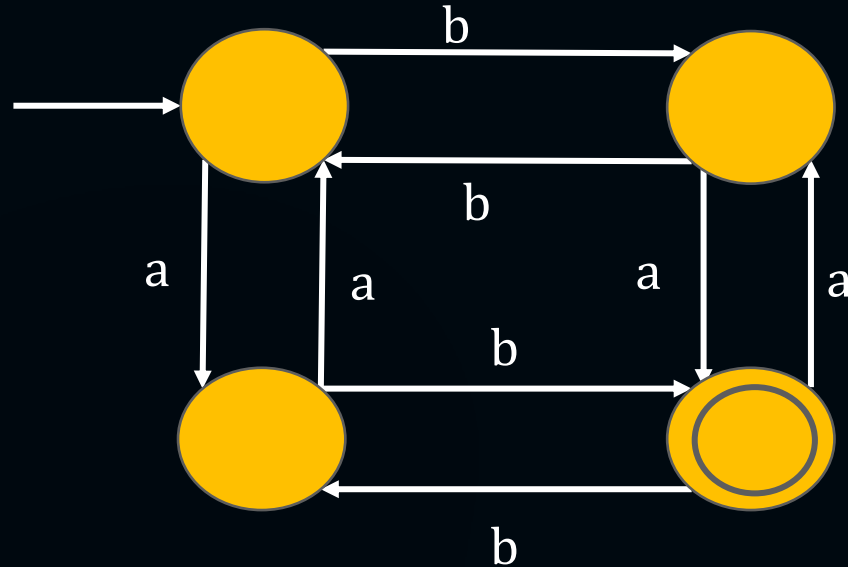
C

a's odd and # b's odd

D

a's even and # b's odd

#Q. Identify language accepted by following DFA



A

a's even and # b's even

B

a's odd and # b's even

C

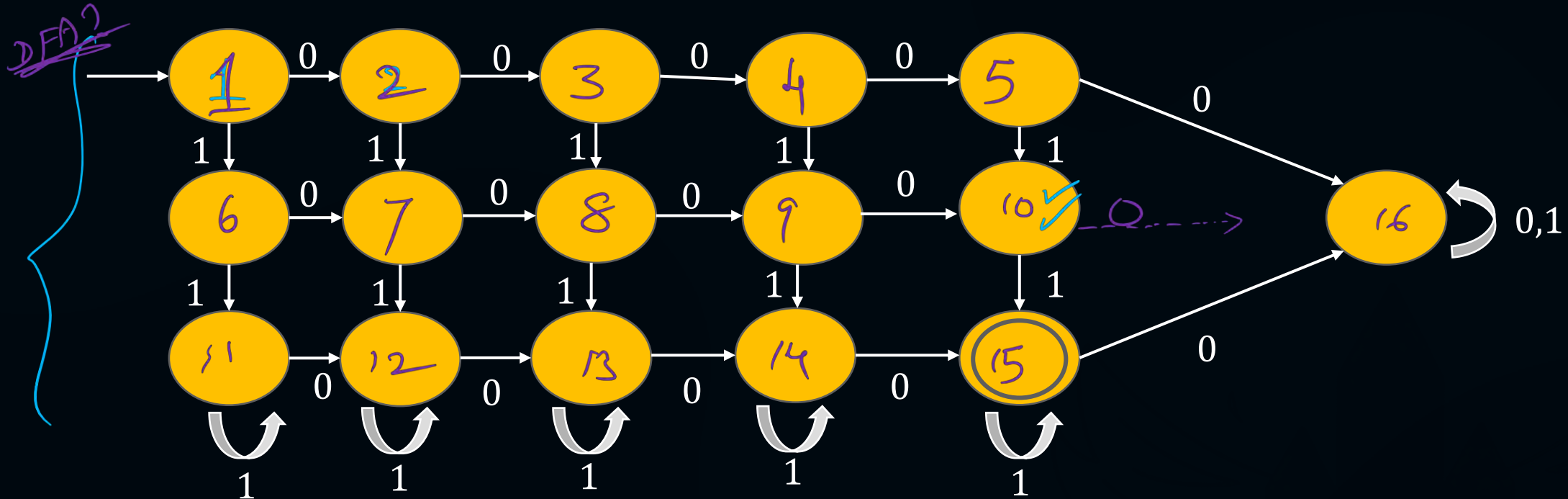
a's odd and # b's odd

D

a's even and # b's odd

$\Sigma = \{0, 1\}$ (Q, Σ , q_0 , F, δ)

#Q. Identify language accepted by following DFA



Set of all strings contains

A

Length of the string atleast 6

B

0's exactly 4 and 1's atleast 3

C

0's atleast 4 and # 1's exactly 2

D

None



Topic : DFA Construction



If comparison exist between symbols of language then DFA is not possible.

i.e If Dependency exist then DFA not possible



Topic : DFA Construction

Finite

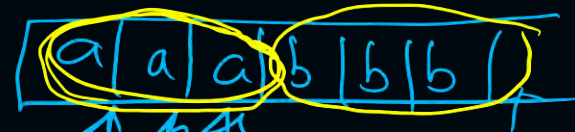
Q Construct DFA for the following Language.

$$L_1 = \{a^n b^n \mid n \geq 1\}$$

$a^{10} b$

dependency exist

DFA not possible



↑ ↑ ↑

q, q_2, q_3

$\{1 = A\}$

any any

$$L_2 = \{a^n b^m \mid n, m \geq 1\} \text{ DFA } \checkmark$$

no dependency

$a^{10} b^{10000}$



Topic : DFA Construction

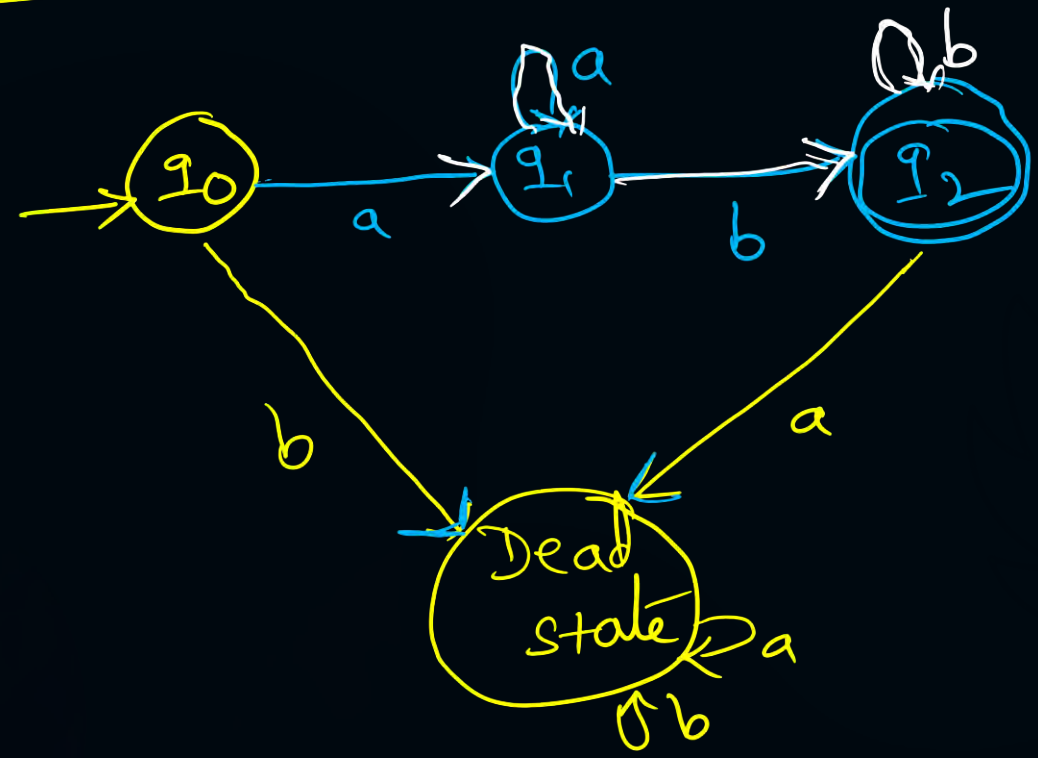
$\{ab, aab, abb, aabb, aaab, \dots\}$

Construct DFA for the following Language.

$L = \{a^n b^m \mid n, m \geq 1\}$

$(Q, \Sigma, q_0, F, \delta)$

DFA?



ba

abaab



Topic : DFA Construction

Construct DFA for the following Language.

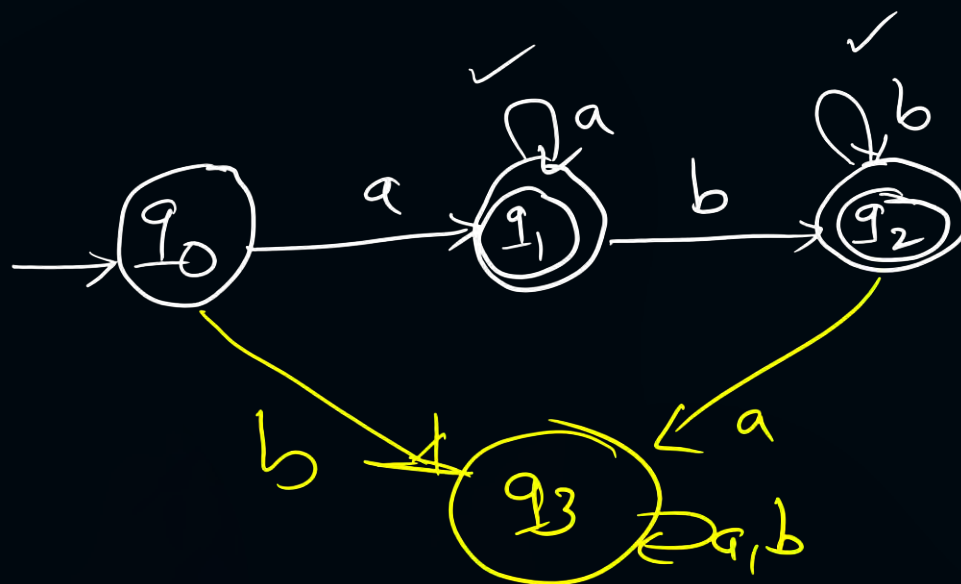
How many states?

~~aba~~

$$L = \{ a^n b^m \mid n \geq 1, m \geq 0 \}$$

@ venkatsir pw

$\{ \overset{\checkmark}{a}, \overset{\checkmark}{a^2}, \overset{\checkmark}{a^3}, \overset{\checkmark}{a^4}, \overset{\checkmark}{a^5}, \dots, \overset{\checkmark}{ab}, \overset{\checkmark}{aab}, \overset{\checkmark}{abb}, \overset{\checkmark}{aaab}, \overset{\checkmark}{abbb}, \dots \}$



4 states



Topic : DFA Construction

Home Work

Construct DFA for the following Language.

$$L = \{a^n b^m \mid n, m \geq 0\} = \left\{ \epsilon, a, a^2, a^3, \dots \right. \\ \left. b, b^2, b^3, \dots \right. \\ \left. ab, aab, \dots \right\}$$



Telegram
@ venkatsispw

THANK - YOU